

1. Administrative & Study Identification

Full Study Title: A Multi Cohort Translational Research Study to Investigate Mechanisms of Resistance to Breast Cancer Therapies
Short Title/Acronym: REMERGE **Protocol Number:** W044977 **NCT Number:** NCT06274515 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Procedure: Tumor Tissue and Blood Draw (Translational Research Study; no investigational drug) **Phase of Development:** Phase 4 **Medical Monitor:** Roche Clinical Operations Lead (global-roche-genentech-trials@gene.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate mechanisms of resistance to anti-breast cancer therapies in tumor and blood samples, specifically assessing how tumors change during treatment to understand why anti-cancer therapies stop working. **Secondary Objectives:** To identify why some tumors do not respond to initial anti-cancer therapies (primary resistance), to compare molecular findings between tumor tissue and blood samples, and to support the development of novel diagnostic tests and tools.

2.2 Background & Rationale To address the gap in understanding acquired and primary resistance to breast cancer treatments. By evaluating matched archival and post-progression tissue alongside blood samples from participants with HER2-positive, HR-positive, or triple-negative breast cancer, the study aims to elucidate the biology of uncontrolled cancer cell growth. The findings will help inform the improvement of current breast cancer therapies and the development of new, targeted therapies in the future.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Translational / Tissue Collection) **Control Type:** Single-arm (Multi-cohort based on receptor status and resistance type) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Estimated 320 participants **Number of Sites:** Global multicenter study (approximately 8 countries) **Estimated Study Start:** 02-Apr-2024 **Estimated Primary Completion:** 30-Sep-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a diagnosis of HER2+, HR+ (cohort R1), or triple-negative breast cancer (cohort T1). 2. Presence of an unequivocally growing tumor lesion (progressive lesion) that is accessible for resection, excision, or core needle biopsy. 3. Availability of an archival tumor tissue (most recent pre-treatment tumor tissue is preferred). 4. Discontinuation of prior anti-cancer treatment should not be longer than 4 weeks from participation in this study, and the participant must have undergone regular disease monitoring (preferably every 3-6 months).

4.2 Exclusion Criteria 1. Any risk factors that increase the risk of complications associated with the procedure to obtain tumor tissue (e.g., bleeding disorders). 2. Participant has already started treatment with a subsequent anti-cancer therapy. 3. Participants whose progressive tumor lesion targeted for biopsy/resection is located in the bone. 4. Discontinuation of the previous treatment was due to a reason other than disease progression.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single intervention visit for a procedural tumor biopsy (or resection) and a matched blood draw. No investigational therapeutic drug is administered. **Route of Administration:** N/A (Biopsy and Venipuncture) **Duration of Treatment:** Total time of participation in the clinical trial is approximately 2-4 weeks, including the biopsy procedure and brief follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Subsequent lines of systemic anti-cancer therapy are prohibited until the required progressive tumor tissue and blood samples have been successfully collected.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Confirmation of Progressive Lesion, Archival Tissue Check Baseline/Intervention Day 0 Tumor Tissue Biopsy/Resection, Matched Blood Draw End of Study Week 2-4 Post-procedure safety follow-up and wound check

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Identification and quantification of mechanisms of resistance to anti-breast cancer

therapies. **Measurement Method:** Multi-omic molecular, genetic, and biomarker analysis of post-progression tumor tissue compared to archival pre-treatment tissue.

7.2 Secondary & Exploratory Endpoints * Concordance of resistance biomarkers between progressive tumor tissue and matched peripheral blood (liquid biopsy) samples. * Characterization of molecular profiles associated with primary vs. acquired non-response to specific anti-cancer therapies.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of procedure-related complications (e.g., localized bleeding, infection, or pain) following the tumor biopsy and blood draw. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor, particularly if related to the biopsy procedure. **Data Monitoring Committee (DMC):** Internal safety and ethics committee oversight by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Biomarker Evaluable Population (participants with successful collection of matched pre-treatment and post-progression samples of adequate quality). **Sample Size Justification:** $N = 320$ allows for robust descriptive multi-cohort analyses across different breast cancer subtypes (HER2+, HR+, TNBC) and resistance profiles (primary vs. acquired). **Handling of Missing Data:** Participants with unanalyzable tissue yields will generally be excluded from matched-pair molecular comparisons.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** On-site and remote monitoring to ensure patient safety during biopsy procedures and secure chain-of-custody for biomarker samples. **Audit Trail:** eCRF locking, tracking of tissue shipment to central translational laboratories, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: W044977 **Registry Number:** NCT06274515 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** REMERGE **Official Regulatory Title:** A Multi Cohort Translational Research Study to Investigate Mechanisms of Resistance to Breast Cancer Therapies

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Cancer (HER2+, HR+, and Triple Negative) **Diagnosis Threshold:** Radiologically or clinically confirmed progressive tumor lesion while on anti-cancer therapy. **Medical Methodology:** Translational Research (Tissue Biopsy and Venipuncture) **Optimization Target:** Discovery of resistance biomarkers to guide future precision oncology therapies.

13. Study Procedures & Methodology

Management Pathway: Identification of progression → Consent → Archival tissue retrieval → Fresh biopsy & blood collection → Initiation of next-line standard of care. **Education Protocol (HCP):** Site training on specialized sample collection, fixation (e.g., FFPE), and central lab shipping procedures. **Education Protocol (Patient):** Informed consent detailing the purpose of tissue donation without direct therapeutic benefit. **Quality Audit Mechanism:** Central laboratory quality control of tumor tissue cellularity, RNA/DNA yield, and sequencing metrics.

14. Observation & Follow-Up Schedule

Total Duration: 2 to 4 follow-up weeks **Onsite Visit Cadence:** 1 main onsite visit for sample collection **Remote Monitoring Frequency:** N/A (Short-term procedural follow-up) **Checklist Review Interval:** End of Study safety check

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				